

Segmenting Public Opinion Without Breaking the Survey Design

Design-weighted latent class analysis with replicate variance estimation

Kevin Linares

Public Opinion Research

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Roadmap

THE PLAN

1. What JPSM trained me to look for
2. Where this work sits in Total Survey Error
3. The problem: trust is a pattern, not a score
4. Why this method, and why not k-means
5. The method, in four moves
6. What we found
7. Naming segments: an anchor and adjust approach
8. What I would bring to our work

The claim in one line

Segmentation of survey data is usually done as if the sample were a simple random sample. It is not, and fixing that changes both the estimates and what we are entitled to say about them.

What I bring back from JPSM

WHERE I TRAINED

Joint Program in Survey Methodology (Maryland, Michigan SRC, Westat). MS in Survey and Data Science, MA in Psychology.

Coursework that shows up here

- ▶ Design-based sampling and estimation
- ▶ Latent variable measurement models
- ▶ Replication variance methods
- ▶ Responsible LLM integration

The two portable habits

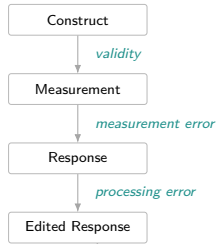
- ▶ Name where error can enter, then decide where to spend effort
- ▶ Everything in code, rendered from source, so a result can be audited a year later

The degree is organised around one framework, **Total Survey Error**. It is the spine of the program and the lens I apply before choosing any method.

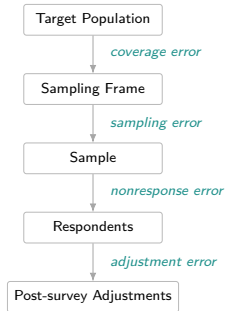
Total Survey Error: the framework the program is built on

GROVES AND COLLEAGUES

MEASUREMENT (the what)



REPRESENTATION (the who)



Survey Statistic

Accuracy is the inverse of mean squared error

THE CORE OF THE FRAMEWORK

$$\text{MSE} = \text{Bias}^2 + \text{Variance}$$

Expanded (Biemer and Lyberg), with each component drawn from one arm or the other:

$$\text{Bias}^2 = \left(\underbrace{\text{specification}}_{\text{meas}} + \underbrace{\text{coverage}}_{\text{rep}} + \underbrace{\text{nonresponse}}_{\text{rep}} + \underbrace{\text{measurement}}_{\text{meas}} + \underbrace{\text{processing}}_{\text{meas}} \right)^2$$

$$\text{Variance} = \underbrace{\text{sampling}}_{\text{rep}} + \underbrace{\text{measurement}}_{\text{meas}} + \underbrace{\text{processing}}_{\text{meas}}$$

Bias and variance do not map onto the two arms. Sampling is representation-side variance, interviewer effects are measurement-side variance. Both arms feed both terms.

MSE is never zero. Even a census carries measurement and processing error. The practical question is never how to eliminate error, it is which component is largest and most tractable under a budget.

Most segmentation work uses only half the map

WHERE THIS PROJECT SITS

Measurement arm

- ▶ **Validity.** Is institutional trust one thing? A summed scale assumes yes. The model tests it.
- ▶ **Measurement error.** Each item is a fallible indicator with its own response probabilities.
- ▶ **Processing error.** Assigning a respondent to a segment is itself an error source. We correct it rather than ignore it.

Representation arm

- ▶ **Sampling error.** Variance comes from the strata and clusters actually used, not a textbook formula.
- ▶ **Adjustment error.** Base weights enter the estimation itself, not a post hoc reweighting of results.
- ▶ **Nonresponse.** Item-partial respondents are scored from what they did answer, above an evidence floor.

Typical practice segments on the measurement arm and reports sample shares. Connecting both arms is what turns a segment share into a **population estimate with defensible uncertainty**.

This is an established method in our field

LCA IN PUBLIC OPINION RESEARCH

- ▶ **Survey measurement error.** Breen (2000, *BJPS*) used LCA to show why surveys underestimate support for extreme parties.
- ▶ **Cross-national attitudes.** Blaydes and Linzer (2008, *World Politics*) characterized respondents across eighteen countries on gender norms, political Islam, and religiosity.
- ▶ **Latin America.** Alvarez, Levin and Núñez (2017, *Journal of Politics*) identified four faces of political participation in Argentina.
- ▶ **Participation repertoires.** Oser, Hooghe and Marien (2013, *PRQ*) tested whether online participation is distinct from offline.
- ▶ **National identity.** Bonikowski and DiMaggio (2016, *ASR*) recovered varieties of American popular nationalism.
- ▶ **Complex survey designs.** Patterson, Dayton and Graubard (2002, *JASA*) fit LCA to NHANES dietary data under the sample design.

The method is not exotic. What is still uncommon is carrying the **sample design** all the way through it.

And specifically for institutional trust

PRIOR TYPOLOGIES

Trust batteries have been segmented before, in several countries and data sources:

- ▶ An Australian national survey recovered four trust classes, from the largely distrusting through the unsure to the broadly trusting.
- ▶ Work on Italian regional data used a latent class measure of trust in others and trust in institutions to map a divide more complex than the usual north and south split.
- ▶ German civic institution data yielded profiles that separate confidence in courts, police, and media from confidence in political institutions.

Two findings recur, and both appear in our results: **trust is not one dimension**, and a substantial group trusts some institutions while rejecting others.

What these studies generally do not report is a design-based interval on any of those group sizes.

The problem: trust is a pattern, not a score

MOTIVATION

The standard move with a trust battery is to sum or average it into a scale.

That assumes everyone who scores a 4 got there the same way. In AmericasBarometer 2023, Mexico they did not:

Two respondents, same average

One trusts the courts and the auditor, distrusts the president.

One trusts the president and the military, distrusts parties and police.

A scale calls them identical.

They are not the same people, they do not respond to the same messages, and they will not move the same way.

Latent class analysis recovers the configuration instead of averaging it away.

Why not k-means?

ALGORITHM VERSUS MODEL

	k-means	Latent class analysis
What it is	An algorithm that partitions this sample	A statistical model of the population
Requires	A distance you choose, usually Euclidean	A likelihood the data imply
Number of groups	Elbow or silhouette, both heuristics	AIC, BIC, entropy: comparable fit
Assignment	Hard. Everyone gets exactly one label	A posterior probability per person
Survey weights	No principled route	Enter the likelihood directly
What you can report	Shares in this file	Population shares with intervals

Three specific failures on these data

- Euclidean distance assumes the gap from 1 to 2 equals the gap

k-means is a fine exploratory tool. The objection is to reporting a cluster share

What the model estimates

METHOD, PART ONE

Each respondent belongs to one unobserved segment. Within a segment, items are answered independently:

$$P(\mathbf{y}_i) = \sum_{k=1}^K \underbrace{\pi_k}_{\text{segment share}} \prod_{j=1}^J \underbrace{\rho_{j y_{ij} k}}_{P(\text{answer}|\text{segment})}$$

What we get out

- ▶ How large each segment is in the population
- ▶ The full response profile that defines it
- ▶ Each respondent's probability of belonging

What it costs

- ▶ The number of segments is a judgment, not an output
- ▶ Local independence is an assumption we have to check

The assumption, and how we check it

LOCAL INDEPENDENCE

The model says: once you know a respondent's segment, their answers to any two items are unrelated. All the correlation between items is explained by segment membership.

If two items stay correlated inside a segment, the model is missing something. We rank every pair by that residual association:

Item A	Item B	Residual association
public_ministry	auditor	5.7
armed_forces	legislature	5.5
justice_system	supreme_court	3.2
police	political_parties	2.8

The top pairs are institutionally adjacent, two audit and prosecution bodies, and two national bodies. We report this as a ranked caution, not a test: under a weighted likelihood there is no valid reference distribution, so an absolute cutoff borrowed from simple random sampling would be misleading.

Why this needed building

THE TOOLING GAP

Tool	Weighted likelihood	Design-based variance	Open
poLCA (R)	no	no	yes
survey (R)	n/a, no mixture models	yes	yes
Mplus, Latent GOLD	yes	linearization	no
baysc (R)	yes, Bayesian	sandwich adjustment	yes
This pipeline	yes	replicate (JKn)	yes

No open frequentist implementation carried a weighted likelihood, design-consistent variance, classification-error correction, and a labelled analyst data product end to end. So the estimation core, the variance machinery, and the profiling correction are custom, and every one of them is validated against a reference implementation.

The method, in four moves

METHOD, PART TWO

1. Weight the likelihood

Every respondent contributes in proportion to the population they represent.

$$\ell_w = \sum_i w_i \log P(\mathbf{y}_i)$$

2. Calibrate the criteria

Base weights do not sum to the sample size, so fit statistics are rescaled before being compared to a sample-size penalty.

3. Replicate the variance

A weighted likelihood gives no usable information matrix. So the whole model is refit on 130 jackknife replicates built from the real strata and clusters.

4. Correct the classification

Assigning people to segments is imperfect. The BCH three-step correction removes the resulting bias before any demographic comparison.

Design: 4 strata, 130 PSUs, 126 degrees of freedom for every interval in this deck.

Move 3: where the uncertainty comes from

REPLICATE VARIANCE

For an ordinary model, standard errors come from the curvature of the likelihood. A **weighted** likelihood is not a real likelihood, so that shortcut is not available and using it anyway understates uncertainty.

Instead we rebuild the estimate many times:

1. Drop one primary sampling unit, reweight its stratum-mates to compensate
2. Refit the entire model on those weights
3. Repeat for all 130 sampling units
4. The spread across those refits is the design-based variance

Two practical details. Each replicate refit is warm-started from the full-sample solution, so the variance measures design uncertainty rather than optimizer wandering. And because segment 1 in one replicate need not be segment 1 in another, replicates are matched back to the full-sample solution before anything is compared.

Move 4: correcting the classification

THE BCH THREE STEP

Respondents are assigned to their most likely segment, but that assignment is imperfect. First we measure exactly how imperfect, using weighted posteriors. Rows are true segments, columns are assigned segments:

$$\mathbf{D} = \begin{bmatrix} 0.94 & 0.04 & 0.02 \\ 0.05 & 0.91 & 0.04 \\ 0.03 & 0.06 & 0.91 \end{bmatrix} \quad D_{ks} = P(\text{assigned } s \mid \text{truly } k)$$

Row one reads: of everyone truly in segment 1, ninety four percent get labelled segment 1 and six percent leak elsewhere. A perfect classifier would be the identity matrix.

Then we invert it. Each respondent's hard label is replaced by the corresponding row of $\mathbf{D}^{-1}$, which un-mixes the leakage. Cross-tabs built on those corrected values are unbiased for true segment composition.

Without this step, every demographic difference is compressed toward the average, and the compression is invisible.

Choosing the number of segments

RESULTS, PART ONE

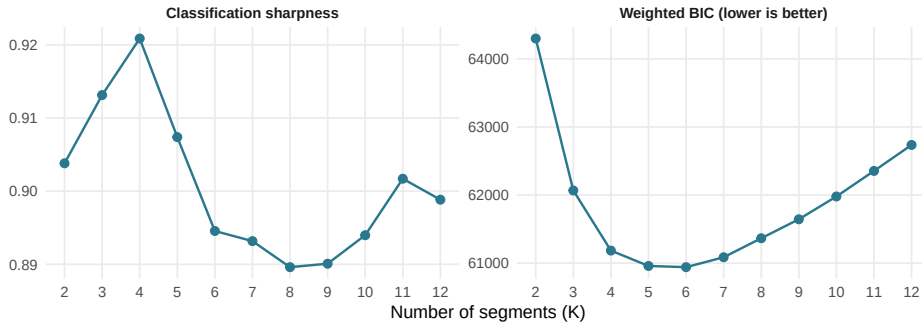


Figure 1

The criterion is nearly flat between five and six segments. We took five, on parsimony, sharper classification, and interpretability. **The software never picks this.** The choice is recorded in configuration and the full evidence is reported so a reader can weigh the alternative.

Five segments, and two of them are configurational

RESULTS, PART TWO

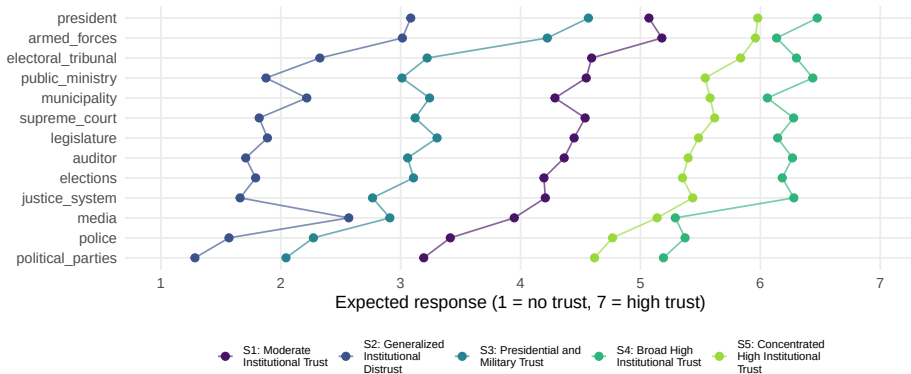


Figure 2

Levels separate three of the segments. The other two cross: same overall trust, opposite shape.

Population size of each segment

RESULTS, PART THREE

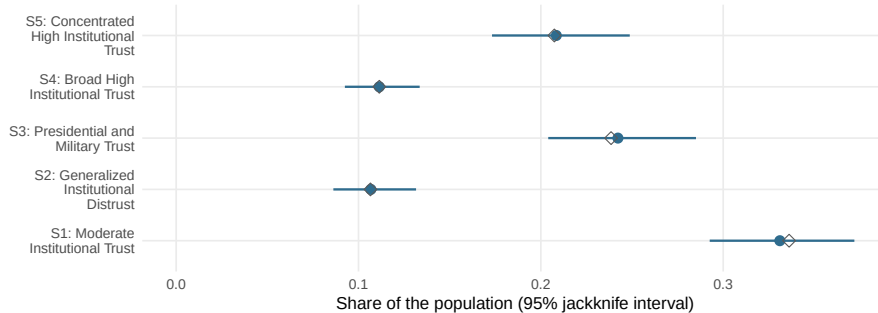


Figure 3

Filled points are design-weighted estimates with jackknife intervals. Open diamonds are what an unweighted analysis would have reported. In this file the weights are nearly constant, so the two agree closely, and that is worth saying out loud. **The intervals are the deliverable:** without them, a three point difference between two segments looks like a finding.

Who belongs to which segment

RESULTS, PART FOUR

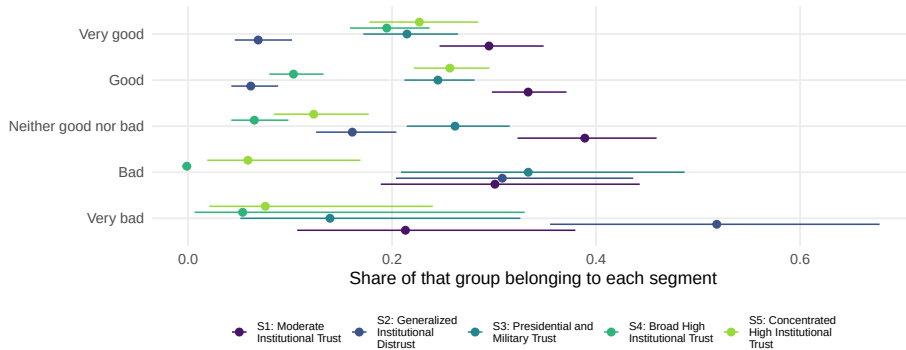


Figure 4

Presidential job approval, corrected for classification error. Composition is strongly ordered, and the intervals tell you which contrasts are real.

Why you can believe the numbers

TRUST CHECKS

Validation. Run at equal weights, this estimator and an independent published implementation must maximize the same function. They agree to 0.00000 on every segment share and 0.00000 on every response probability.

- ▶ **One verdict, one place.** The validation result is written to disk by the estimation script and read by every downstream document, so no two outputs can disagree.
- ▶ **Frozen decisions.** The number of segments and the segment names live in files under version control, not in someone's memory.
- ▶ **Rendered from source.** Every figure in this deck was generated from the stored model objects at compile time. Nothing was pasted.

1618 of 1622 respondents received a segment, including 188 who skipped items but answered enough to be scored.

Naming the segments is the real bottleneck

FROM MEASUREMENT TO LANGUAGE

The model returns 5 segments as 455 conditional response probabilities. Someone still has to read that grid and say what each segment *is*.

Done by hand, this is

- ▶ Slow, and it is the step that delays every deliverable
- ▶ Inconsistent across analysts and across waves
- ▶ Undocumented: the reasoning lives in someone's head

But it cannot simply be automated

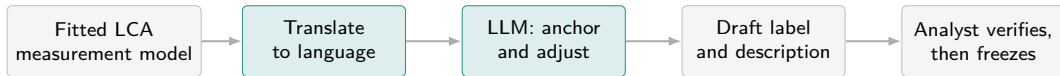
- ▶ A name is an interpretive claim about a population
- ▶ A fluent wrong label is worse than no label, because it propagates into every downstream table

So the model drafts and the analyst decides. The design question is how to constrain the draft so that verifying it is faster than writing it.

Anchor and adjust: the same pattern as my capstone

ARCHITECTURE

In my JPSM capstone (Latent-to-Language) the pattern was: fit a measurement model, translate its **conclusions** into natural language, then ask the model to adjust from that anchor rather than reason from scratch. The same architecture applies here.



Translation, not raw parameters. The model never sees a probability matrix. It sees each item's actual question wording and each probability against the answer text the respondent saw:

police "Confianza en la policía nacional" \ $P(\text{Nada})=0.08$, $P(2)=0.17$, $P(3)=0.29$, $P(4)=0.26$,
 $P(5)=0.14$, $P(6)=0.04$, $P(\text{Mucho})=0.02$

The instrument, verbatim

PERSONA AND RULES

SYSTEM PROMPT (the persona)

You are a senior survey methodologist who reads latent class analysis (LCA) measurement models. In this work each latent class is called a SEGMENT; that is a word-choice preference and the statistical object is unchanged. Each segment is described only by its item-response probabilities: for every survey item, the probability that a member of that segment gives each answer. A segment leans toward the answers with high probability. You interpret a segment strictly from these probabilities and the item wording, never from outside assumptions.

RULES (sent with every call)

1. Use only the response probabilities and item wording shown. The survey context only clarifies what the items refer to; attribute nothing to the segment that the probabilities do not show.
2. Anchor every statement to the high-probability answers of this segment.
3. If the profile is diffuse (no clear high-probability answers), say so.
4. Return only valid JSON: no prose before or after, no markdown fences.

Rule 2 is the anchor. Rule 3 is calibrated hedging: the model is required to admit when a segment has no clear shape rather than invent one. Rule 1 fences the survey context to resolving what an institution *is*, so no outside knowledge can enter the interpretation.

What keeps it honest

GUARDRAILS

Reproducible by construction

- ▶ Decoding is deterministic: temperature zero, fixed seed
- ▶ One isolated call per segment. Joint prompts empirically confused near-neighbour segments
- ▶ Output is JSON, validated on arrival, never free prose
- ▶ Labels are frozen to a CSV under version control. That file, not a seed, is the reproducibility mechanism

Audited, not trusted

- ▶ A second bounded call harmonizes labels only when two collide, so most runs make no extra call at all
- ▶ The original draft is retained alongside any edit, so every change is visible in the frozen file
- ▶ The whole instrument is certified against planted test cases: near-neighbour segments, a deliberately diffuse segment, and a fictional survey context to test for context leakage
- ▶ Any edit to the prompt obliges a re-certification run

The prompt is treated as a measurement instrument: versioned, tested, and re-validated when changed.

Where the line sits

A DELIBERATE BOUNDARY

The model drafts

- ▶ A label and description per segment
- ▶ From the estimated probabilities only
- ▶ Into a file the analyst edits and owns

The model never decides

- ▶ How many segments there are
- ▶ Which items to keep or drop
- ▶ What the results mean
- ▶ Anything that appears as a finding

The rule I would apply to any AI use in our shop:

a model may draft configuration, never a decision and never a claim about results.

Every substantive statement in this briefing rests on the estimated probabilities and their intervals, not on anything a language model wrote.

What I would bring to our work

NEXT STEPS

Ready now

- ▶ Segmentation of any categorical battery with population estimates and design-based intervals
- ▶ Design-based inference review on analyses we already publish
- ▶ Reproducible pipelines that render from source

Where I would like to go next

- ▶ Package the workflow so a new study is a configuration file, not a copied folder
- ▶ Extend the profiling correction to trends across waves
- ▶ Evaluate language model assistance where it is auditable

The offer: bring me a battery of categorical items and its design variables, and I will tell you how many groups are really in there, how big they are in the population, and how sure we are.

Questions

Thank you

Methods write-up, code, and the full set of diagnostics available on request.

Backup slides follow: full response profiles, item screening, references.

Backup: full response profiles

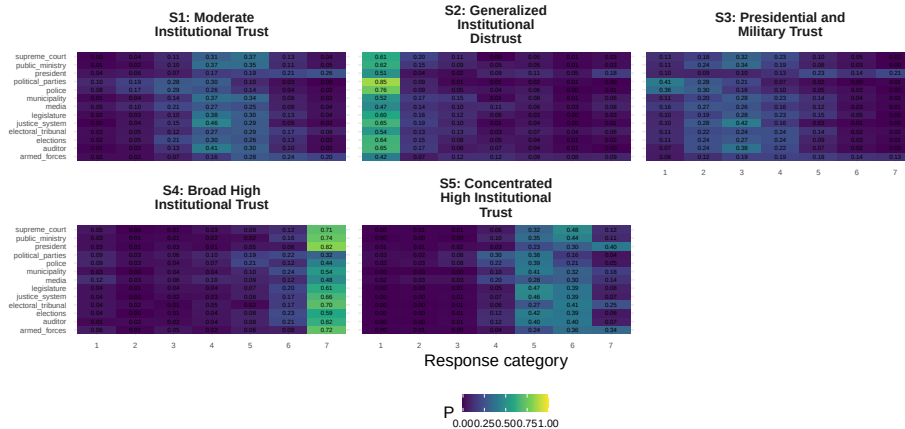


Figure 5

Backup: which items carry the segmentation

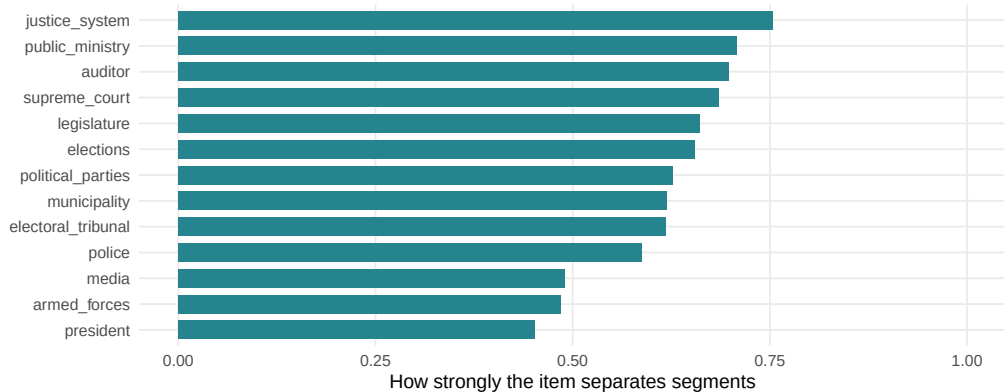


Figure 6

Every item earns its place. The two lowest, the president and the armed forces, are the institutions on which otherwise different segments agree, which is itself the finding.

Backup: references

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Method

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